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Research report on: blockchain security

Outline

Introduction

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Literature Review

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Methodology

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Discussion/Findings

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Research Gaps & Future Work

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Conclusions

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Literature Synthesis

Sources:

[Kim, 2021] Advanced Drone Swarm Security by Using Blockchain Governance Game

Abstract: This paper presents a general cost model for two-person mixed-strategy games, where player H has two strategies at the observation moment. The cost matrix is composed as follows: NotBuers, Burst Regular Safe. The cost matrix is derived based on given conditions, where NotBuers is the cost of the game and Burst Regular Safe is the general cost matrix at the prior time to be composed as follows: NotBuers, Burst Regular Safe. The paper's main contribution is to derive the cost matrix, which enables the general cost matrix at the prior time to be calculated. The paper presents the methodology for estimating the cost of a game, and the results indicate that NotBuers, Burst Regular Safe is the most profitable strategy. The paper identifies three key findings:

1. The cost model for two-person mixed-strategy games.

2. The methodology for estimating the cost of a game.

3. The findings of the paper.

[Panait, 2020] Identity Management on Blockchain -- Privacy and Security Aspets

Abstract: This paper aims to provide an overview of the current status in the field of identity management on blockchain, including privacy and security aspects. This paper discusses the need for a better understanding of these aspects to ensure their successful implementation. The paper also identifies existing chaullenges and new directions for research. Key findings: The paper presents a brief overview of the available literature in the field of identity management on blockchain, highlighting the chalee nges and limitations of such solutions. It also identifies several key finding areas, including the need for secure and user-centric identity management solutions, the importance of privacy and data protection, and the need for research on the role of blockchain in identity management.

[Wang, 2019] ChaingSplitter: Towards Blockchain-based Industrial IoT Architecture for Supporting Hierarchical Storage

Abstract: This paper proposes a blockchain-based Industrial Internet of Things (IIoT) architecture that provides immutable and veri?able services for large-scale IIoT systems to build large-scale systems to

connect numerous heterogeneous devices and systems into the Internet. This paper proposes a hierarchical blockchain storage structure that allows the majority of the blockchain to be stored in the cloud while the most recent blocks are stored in the overlay network of the individual IIoT networks. The proposed hierarchical blockchain storage structure allows the majority of the blockchain to be stored in the cloud, while the most recent blocks are stored in the overlay network of the individual IIoT networks.

[Wang, 2020] Blockchain-based Industrial IoT (IIoT) Network Architecture for Supporting Hierarchical Storage

Abstract: In this paper, the authors propose a blockchain-based IIoT network architecture that provides immutable and veri?able services for large-scale IIoT systems. This paper proposes a hierarchical blockchain storage structure that allows the majority of the blockchain to be stored in the cloud while the most recent blocks are stored in the overlay network of the individual IIoT networks. The proposed hierarchical blockchain storage structure allows the majority of the blockchain to be stored in the cloud, while the most recent blocks are stored in the overlay network of the individual IIoT networks, providing a scalable and efficient IIoT network architecture. The proposed hierarchical blockchain storage structure enables a scalable and efficient IIoT network architecture.

[Panait, 2021] Advanced Drone Swarm Security by Using Blockchain Governance Game

Abstract: The paper presents a general cost model for two-person mixed-strategy games, where player H has two strategies at the observation moment. The cost matrix is composed as follows: NotBuers, Burst Regular Safe. The cost matrix is derived based on given conditions, where NotBuers is the cost of the game and Burst Regular Safe is the general cost matrix at the prior time to be composed as follows: NotBuers, Burst Regular Safe. The paper's main contribution is to derive the cost matrix, which enables the general cost matrix at the prior time to be calculated. The paper presents the methodology for estimating the cost of a game, and the results indicate that NotBuers, Burst Regular Safe is the most profitable strategy. The paper identifies three key findings:

1. The cost model for two-person mixed-strategy games.

2. The methodology for estimating the cost of a game.

3. The findings of the paper.

Research Gaps & Future Work

1. Determine the optimal solution to the problem of secure data transmission between devices in the IIoT network
2. Identify the most effective ways to manage and protect the blockchain data stored in the IIoT network
3. Develop a cost-effective solution for the maintenance and management of the IIoT network
4. Develop a scalable and efficient IIoT network architecture using blockchain technology
5. Improve the security and privacy of IoT devices by leveraging blockchain technology
6. Identify and implement solutions for the prevention of IoT network attacks and data breaches

These research gaps and open questions will provide a foundation for further research and development in the field of IIoT and blockchain.

Paper Summaries

Advanced Drone Swarm Security by Using Blockchain Governance Game

Song-Kyoo Kim (2021)

The paper presents a general cost model for two-person mixed-strategy games, where player H has two strategies at the observation moment. It considers both the normal form of games and the general cost matrix at the prior time to be burst. The paper includes a general cost matrix at the prior time to be composed as follows: NotBurst Burst Regular Safe. The cost matrix is derived based on the given conditions, where NotBursst Burst Regular Safe represents the cost of the game. The paper's main contribution is to derive the cost matrix, which enables the general cost matrix at the prior time to be calculated. Moreover, the paper presents the methodology for estimating the cost of a game, and the results indicate that NotBursst Burst Regular Safe is the most profitable strategy. Finally, the paper lists 3-5 key finding in bullet points, each with a short summary.

Identity Management on Blockchain -- Privacy and Security Aspects

Andreea-Elena Panait, Ruxandra F. Olimid, Alin Stefanescu (2020)

Paper Title: Identity Management on Blockchain -- Privacy and Security Aspects Keywords: Digital identity, identity management, blockchain, self-sovereign identity. 1. Motivaion and Contribution The paper aims to provide an overview of the current status in the field of identity management on blockchain, including privacy and security aspects. Motivaioned by the fact that several studies suggest that blockchain technology can be an effective solution for digital identity management, the paper discusses the need for a better understanding of these aspects to ensure their successful implementation. The paper also identifies existing challenges and new directions for research. Key findings: The paper presents a brief overview of the available literature in the field of identity management on blockchain, highlighting the challenges and limitations of such solutions. It also identifies several key finding areas, including the need for secure and user-centric identity management solutions, the importance of privacy and data protection, and the need for research on the role of blockchain in identity management. 2. Related Work The paper provides a comprehensive overview of identity management on blockchain, including user-centric identity, authentication solutions, and digital identity management in the context of IoT. The paper also discusses self-sovereign solutions, the implementation of which with respect to the principles of the principles of self-sovereign identity. 3. Summary and Bullet Points The paper provides a brief summary of the core contribution and methodology, highlighting the key findings and key findings areas. The bullet points provide short summary and highlight key findings as bullet points, each on its own line prefixed with '- '.

ChainSplitter: Towards Blockchain-based Industrial IoT Architecture for Supporting Hierarchical Storage

Gang Wang, Zhijie Jerry Shi, Mark Nixon, Song Han (2019)

Title: ChaingSplitter: Towards Blockchain-based Industrial IoT Architecture for Supporting Hierarchical Storage Abstract: The fast-evolving Industrial Internet of Things (IIoT) technologies offer a promising opportunity to build large-scale systems to connect numerous heterogeneous devices and systems into the Internet, but the required storage space poses a graft challenge to resource-constrained IIoT infrastructures. In this paper, the authors propose a blockchain-based IIoT architecture that provides immutable and veri?able services for large-scale IIoT systems by storing data in the cloud, while maintaining records in local IIoT networks. The proposed hierarchical blockchain storage structure allows the majority of the blockchain to be stored in the cloud, while the most recent blocks are stored in the overlay network of the individual IIoT networks. To ensure ef?ciency of the proposed hierarchical blockchain storage, the authors provide a case study and showcase the efficiency of the proposed hierarchical blockchain storage in a practical Industrial IoT

case. The authors propose a novel blockchain storage structure, called ChaingSplitter, to store the blocks in a hierarchical manner. This contribution, methodology, and results provide a novel solution for supporting immutable and veri?able services in large-scale IIoT systems by leveraging the advantages of blockchain technology. The authors' findings include the ef?ciency and scalability of the proposed hierarchical blockchain storage, which can support large-scale IIoT systems.

Bibliography

Kim, S. (2021). Advanced Drone Swarm Security by Using Blockchain Governance Game. arXiv. https://doi.org/10.3390/math10183338

Panait, A., Olimid, R. F., & Stefanescu, A. (2020). Identity Management on Blockchain -- Privacy and Security Aspects. arXiv. http://arxiv.org/abs/2004.13107v1

Wang, G., Shi, Z. J., Nixon, M., & Han, S. (2019). ChainSplitter: Towards Blockchain-based Industrial IoT Architecture for Supporting Hierarchical Storage. arXiv. https://doi.org/10.1109/Blockchain.2019.00030